

CO₂ and Photosynthesis

Pupil sheet

Name: _____

Date: _____

Safety: The box is not waterproof: never water the plant in the jar. Wash your hands after touching soil.

1. What I see on the screen

On the screen: **CO₂** (ppm), **Light** (lx), **Temperature** (°C), **Humidity** (%) and the **CO₂ rate**, a gradient with a $\pm$. **If the lamp heats the jar, we have changed two factors at once.**

I read the gradient, not the value. After every change I wait **3 minutes**, then read the gradient over at least **3–5 minutes**. If the CO₂ falls towards **200 ppm** or the humidity passes **90%**, I air the jar for a minute and repeat the step.

In light I measure *net* photosynthesis (photosynthesis minus respiration). **Gross** = net + the respiration from darkness. A gradient differs from zero only if it is clearly larger than **twice its $\pm$** .

2. What I do on my own

First I predict. In bright light the CO₂ will: ☐ $\uparrow$ ☐ $\downarrow$ ☐ $\rightarrow$ In darkness it will: ☐ $\uparrow$ ☐ $\downarrow$ ☐ $\rightarrow$

Is there a light level at which the line stays flat? ☐ yes ☐ no

because _____

The rig: plant _____ jar _____ L lamp at _____ cm

The basic experiments. Example: the app shows -10 ± 1 ppm/min over 5 min. Check by hand: $820 \rightarrow 770$ ppm in 5 min $\Rightarrow (770 - 820)/5 = -10$ ppm/min.

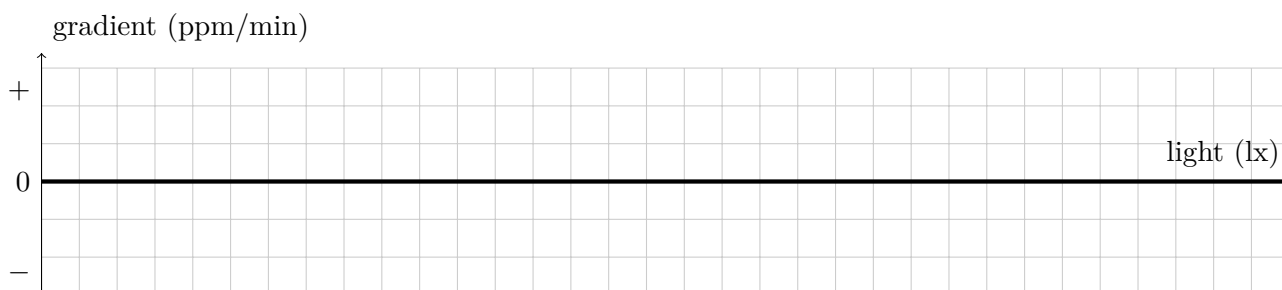
Condition	Light (lx)	Minutes	Gradient (ppm/min, with $\pm$)
Empty jar (seal test)			
Pot without plant (soil)			
Plant, bright light			
Plant, darkness	0		

I subtract the soil's gradient. Net photosynthesis = _____ respiration = _____ gross = _____ ppm/min

The intensity series. At every step I check the temperature and press **Capture**.

Shading	Light (lx)	Gradient (ppm/min, with ±)	<i>T</i> (°C)
No sheet			
One sheet			
Two sheets			
Three sheets			
Darkness	0		

The graph. I plot the points. **The compensation point is where the curve crosses the thick zero line.**



Compensation point $\approx$ _____ lx (the app: _____ lx). Check: the value at 0 lx should be the respiration rate.

3. The conclusion I write

I predicted that, and the data show that _____

When the line was flat, what was happening in the leaves was _____

I cannot conclude anything about the planet's forests because _____

Exit ticket

What is happening in the leaves when the CO₂ line is flat?
